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Skip-Softmax + FP8-SAGE Attention Backend

Integration RFC — composable trtllm-gen attention for Diffusion Transformers

via FlashInfer: block sparsity (BLASST) × SAGE FP8 attention × ModelOpt GEMM quant

Algorithm: BLASST (NVIDIA DevTech, MLSys'26) · Targets: Wan 2.2 · Hunyuan Video · Cosmos 3

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Motivation

Why now, and what we're proposing

- Video DiT inference is attention-bound at long context (Wan 2.2: 81 frames × 720p → >75k tokens).
- BLASST Skip-Softmax skips negligible attention blocks inside the kernel — no training, no weight change.
- It composes with SAGE (FP8 attention) and ModelOpt GEMM quant. On Wan 2.2 / B200: up to 1.40× E2E.
- Blackwell trtllm-gen kernels are already exposed via FlashInfer; kernel parity vs TRT-LLM RC20 validated on B300.

1.40×

NVFP4 + SAGE + Skip vs BF16 dense

Proposal: a composable diffusion attention backend trtllm with sage + skip_softmax features.

Algorithm Overview

BLASST — skip a block when it can't matter

```
m̃_j = rowmax(Q·K_j^T)        # block max
m   = max(m, m̃_j)            # running max
if  m̃_j − m < ln(λ):
        continue              # SKIP block
P   = exp(Q·K_j^T − m)
O += P · V_j                  # else accumulate
```

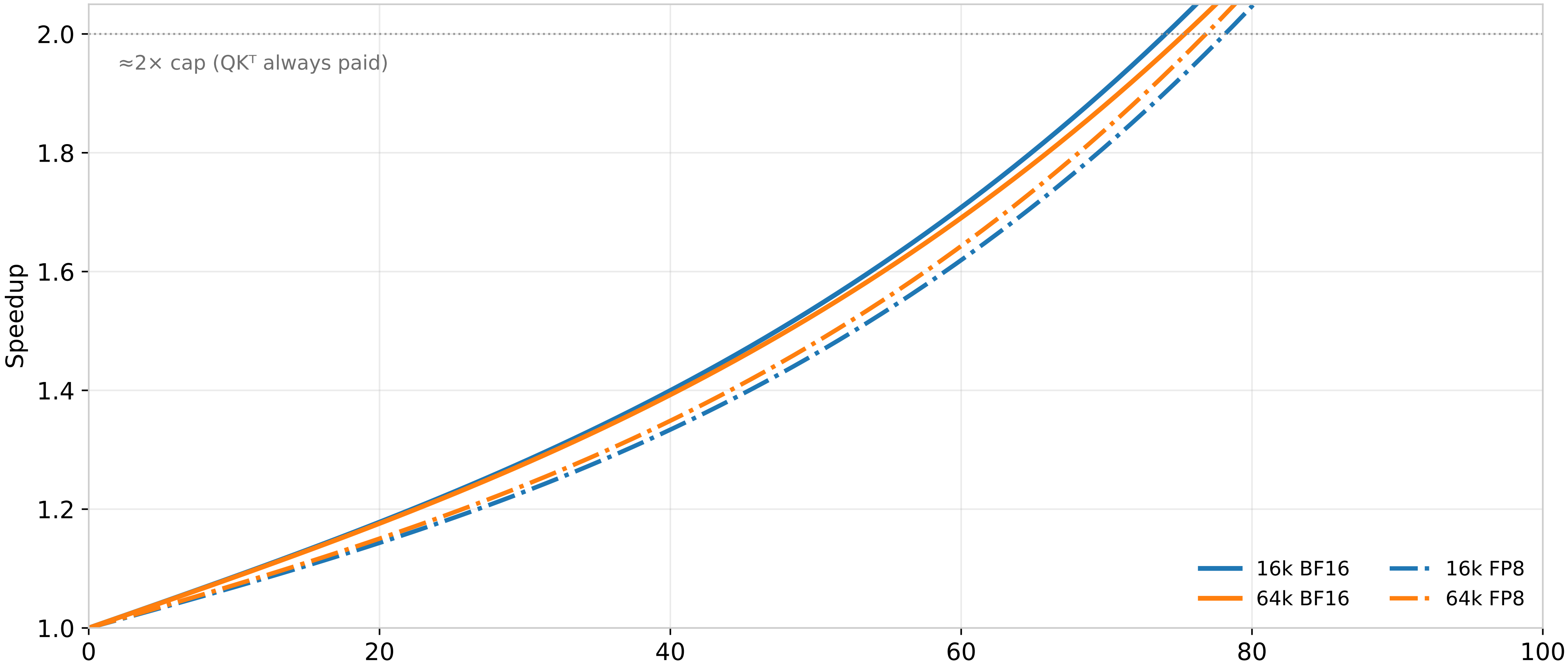
- Skip test uses the running max → monotone, so a skip is never retroactively wrong (no rescale).
- Skips the P·V GEMM + V-load; still pays QK^T → kernel speedup capped ~2×.
- Threshold $\lambda = a/L$ (context-length normalized); a is calibrated, not analytic.

per-block decision along the KV axis:

KEEP	SKIP	SKIP	KEEP	SKIP	KEEP

Kernel-level Speedup

Blackwell (B200) prefill — speedup vs achieved sparsity



Roughly linear: ~1.3× @50%, ~1.6× @80%. BF16 gains more than FP8 (un-skippable QK^T is a bigger fraction under FP8).

Where It Plugs In

Reuse vLLM-Omni's diffusion attention stack + upstream precedent

vLLM-Omni already has the seams

- DiffusionAttentionBackendEnum (flat backend registry)
- attention/selector.py — hardware/shape-gated pick + fallback
- attention/layer.py :: Attention(q, k, v, md)
- ModelOpt FP8/NVFP4 checkpoint loader — shipped (#5076, #5087)

Mirror how upstream vLLM exposes trtllm

- Backend = FLASHINFER; trtllm-gen is a kernel path inside it (TRTLLMPrefill / TRTLLMDecode).
- Gated by supports_/can_use_/force_use_trtllm_attention().
- Copy the pattern that matters: auto-detect · force flag · graceful fallback — not the name.

Decision: expose diffusion-side name trtllm → resolves to FlashInfer trtllm-gen; sage / skip_softmax are features under it.

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PRINCIPLE

API Design

Split the kernel from the DiT policy

Pipeline / runtime (owns the DiT policy)

denoise step + total steps → gate: dense if t > disabled_until → factor = $\theta(\text{sparsity}, L) \cdot L$ → md.skip_softmax_factor

Backend (pure kernel adapter — no diffusion knowledge)

bf16 q,k,v → SAGE quant → fp8 + scales → pack ragged (cu_seqlens) → trtllm_ragged_attention → bf16 out

BLASST is the kernel; the timestep gate is integration. Keep them in different layers so the backend stays a pure adapter.

API Design

One backend flag + two dials users understand

```
stages:
  - stage_id: 0
    attention_backend: trtllm      # trtllm-gen
    trtllm:
      sage: true                   # FP8-SAGE
      skip_softmax:
        sparsity: 0.65             # speed↔fidelity
        disabled_until_timestep: 0.86
        calibration: skip_calib.json
```

```
vllm serve my/dit --omni \
  --attention-backend trtllm \
  --attention-config '{"sage":true,
    "skip_softmax":{"sparsity":0.65}}'
```

The only two concepts a user needs

- sparsity (0 → ~0.9): speed↔fidelity dial. 0 = dense.
- disabled_until_timestep (0–1): keep early denoise steps dense — protects visual fidelity.
- Hidden: FP8/SAGE scales, block layouts, λ=a/L, calibration lookup.

Per-request override

```
{"num_inference_steps":50,
 "extra_body":{"skip_softmax_sparsity":0.6}}
```

API Design

Config schema + graceful fallback

```
@dataclass
class SkipSoftmaxConfig:
    sparsity: float | None = None
    calibration: str | None = None
    disabled_until_timestep: float = 0
    theta: float | None = None    # advanced

@dataclass
class TrtllmAttnConfig:
    sage: bool = True
    sage_block = (1,4,0,1)   # (Q,K,P,V)
    skip_softmax: SkipSoftmaxConfig | None
```

Selector: only run the validated envelope

- SM100+ (B200/B300); head_dim=128; dense, non-causal; MHA (q_heads=kv_heads).
- Outside it → auto-fall-back to flash-attn with a one-line log. No silent no-ops, no crashes.
- No calibration and no theta → skip disabled (degrade to dense, never guess).
- use_trtllm=false / sparsity:0 = hard off switch.

ModelOpt in the Loop

Offline once; two jobs, not one — and they compose

Job 1 · GEMM quant

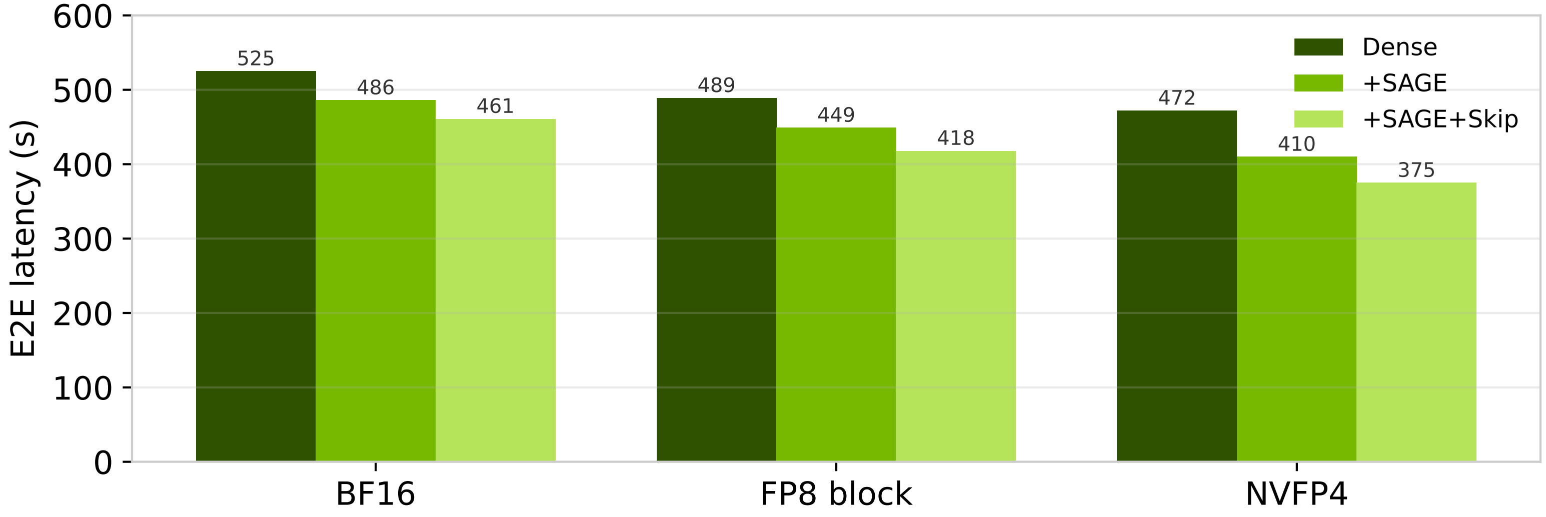
Quantize linear weights → FP8/NVFP4, scales baked into the checkpoint. Loaded by the existing adapter (#5076/#5087).

Job 2 · Skip calibration

Fit sparsity↔θ map + recommend disabled_until_timestep. No weight change → skip_calib.json.

NOT ModelOpt · SAGE

FP8-SAGE attention scales are dynamic/runtime, per-block — computed each forward. No calibration.



Wan 2.2 T2V-A14B · 720p · 81f · 50 steps · B200 · target_sparsity=0.65, disabled_until=0.86

GEMM	Dense	+SAGE	+Skip
BF16	1.00×	1.08×	1.14×
FP8	1.07×	1.17×	1.26×
NVFP4	1.11×	1.28×	1.40×

Skip is the smallest contributor (~+5–9% over SAGE). Most of the win is GEMM-quant × SAGE.

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Test Plan

Kernel parity + unit + fallback

Kernel parity (harness exists)

- FI trtllm-gen vs pinned TRT-LLM RC20, byte-identical FP8 inputs.
- Gate: rel-L2 ≤ 2e-3, cos ≥ 0.999, Skip-delta ≤ max(2e-4, 0.05·R_ref).
- Partial-K tails (S257/258/259) + mod-4 cumulative offsets.
- B300 result: max rel-L2 = 1.36e-5 ✓

Unit / regression (in-tree)

- Reject negative / NaN / inf skip factors.
- None == 0.0 disable equivalence (≤1e-6).
- Active factor materially changes output; still finite.
- Unsupported SM / head_dim / causal → falls back, no exception.
- Uncalibrated model → skip disabled (dense), not garbage.

END-TO-END

Test Plan

Prove it on Wan 2.2 · Hunyuan Video · Cosmos 3

Model	Task / shape	Perf goal	Fidelity gate
Wan 2.2 A14B	T2V 720p · 81f · 50 steps	reproduce ≥1.3× (NVFP4+SAGE+Skip)	LPIPS-low5 + VBench vs BF16-dense
Hunyuan Video	T2V 720p · long context	≥1.25× E2E, attn kernel ≥1.4×	LPIPS + human spot-check
Cosmos 3	world-model / video DiT	≥1.2× E2E; no OOM at 6 GiB KV	PSNR/LPIPS vs FP8-dense

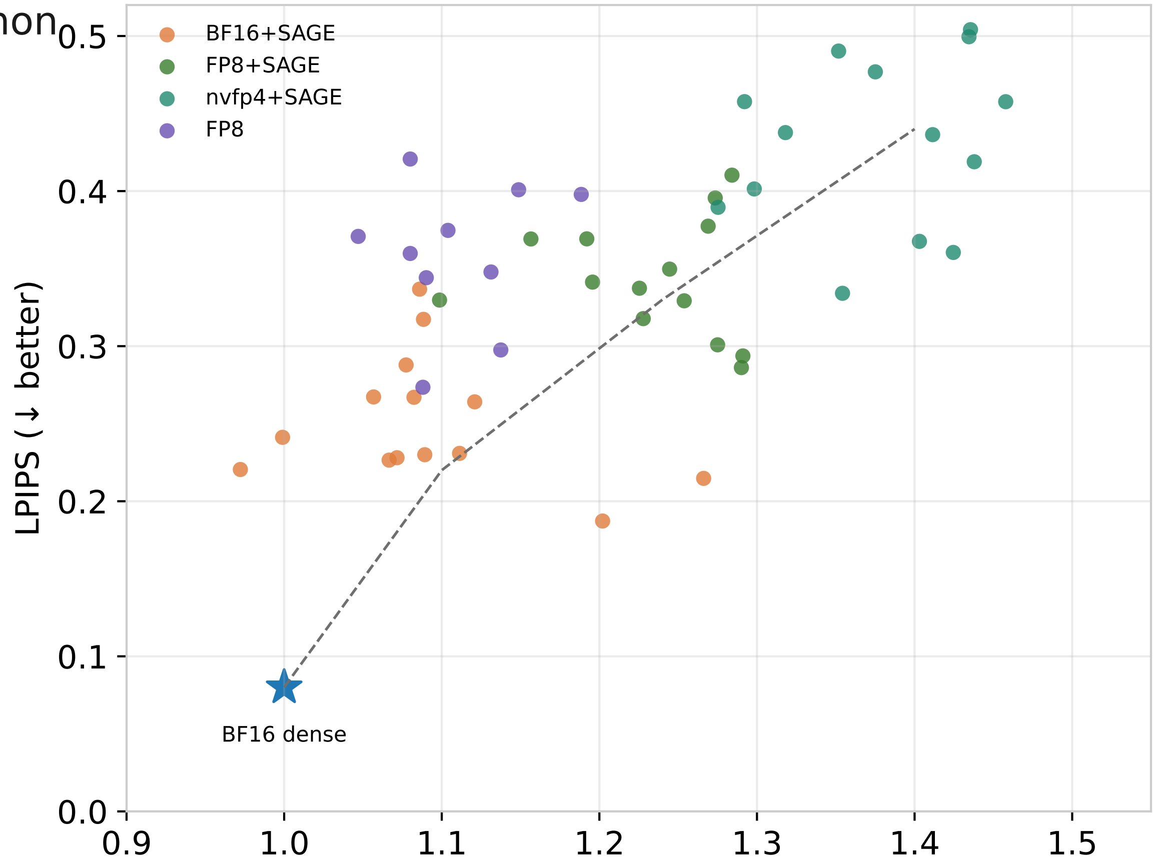
Sweep to build the Speedup–Fidelity Pareto

- Per model: sparsity ∈ {0, 0.3, 0.5, 0.65, 0.8} × disabled_until_timestep ∈ {0.7, 0.86, 0.95}.
- On-the-fly SAGE ≠ the exact-input kernel test → E2E gets its own quality gate.
- Publish the frontier + the recommended default per model.

Metrics & Gates

What we measure, and the bars to pass

- E2E latency & speedup vs BF16 compiled-dense (common baseline).
- Attention kernel time + achieved sparsity (does θ realize target).
- Peak VRAM within KV budget (~13 GiB).
- Fidelity: LPIPS-low5 / PSNR / VBench vs dense reference.
- Perf gate: FI vs TRT median ≤1.02×, p95 ≤1.05×.
- CI: parity+unit per PR (L1/L2); E2E sweeps nightly (L4) on a Blackwell runner.



SAGE variants own the frontier; quality-safe ≤ 1.25×

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Rollout & The Ask

Phased, honest about limits

Phases

- P0 — trtllm backend + selector gating + flash-attn fallback (no skip yet).
- P1 — SAGE FP8 attention path + kernel-parity CI.
- P2 — Skip-Softmax + skip_calib.json + timestep gate.
- P3 — per-model calibration + defaults + docs (Wan / Hunyuan / Cosmos 3).

Risks / open

- Blackwell-only (SM100+); everything else falls back.
- On-the-fly SAGE quant ≠ validated exact-input → own E2E quality gate.
- Per-model calibration artifact ownership + versioning.
- Skip is the smallest win → opt-in per model, not default-on.

Ask: sign-off to build P0–P1 (register trtllm backend + SAGE + kernel-parity CI). Lead E2E bring-up: Wan 2.2.

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